

ISHAQ SHAIK

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EDUCATION

Arizona State University

Master of Science in Computer Science • GPA: 3.8/4.00

- Relevant Coursework: Statistical Machine Learning, Artificial Intelligence, Data Processing at Scale, Semantic Web Mining

Tempe, AZ

Aug. 2024 – May 2026

Ramaiah Institute of Technology

Bachelor of Engineering in Information Science

Bangalore, India

Aug. 2019 – May 2023

TECHNICAL SKILLS

Languages Python, SQL, Java, C++, JavaScript, Bash

ML & AI PyTorch, TensorFlow, Scikit-learn, Gradient Boosting, Random Forest, Logistic Regression, Transformers

Data & Analytics Pandas, NumPy, Matplotlib, Seaborn, PostgreSQL, DuckDB, Redis

Cloud & Tools AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, GitHub, CI/CD, Linux, SLURM

EXPERIENCE

Arizona State University

Graduate Student Assistant

May 2025 – May 2026

Tempe, AZ

- Deployed a retrieval-augmented generation chatbot serving **5,000+** users at 1.2s latency with AWS Lambda inference.
- Cut multi-turn compute costs by **40%** via context summarization, payload packing, and secure session state management.
- Scaled an AWS Lambda + RDS ML pipeline for **300k+** events at **99.9%** uptime, automating ingestion from monthly to daily.

Extreme Networks

Software Engineer

Jul. 2023 – Jun. 2024

Bangalore, India

- Reduced inter-service latency by **25%** across 1M+ daily requests by migrating Java REST APIs to gRPC + Protobuf.
- Delivered **8** Spring Boot data APIs across 20+ NFL/MLB stadium sites, validating 70+ edge cases for flagship hardware.

Nbyula

Software Engineer Intern

Mar. 2023 – May 2023

Bangalore, India

- Cut image load times by **84%** (2.5s to 400ms) via CloudFront edge caching and server-side responsive cropping.
- Raised chat DAU by **32%** (3.2K to 4.2K) by fixing Redis cache invalidation bugs and optimizing React render paths.

LoadShare Networks

Software Developer Intern

Sep. 2022 – Dec. 2022

Bangalore, India

- Eliminated manual SQL ops saving **15 min/day** by building a Spring Boot Slack bot with automated query execution.
- Accelerated deployment by **75%** (40 to 10 min) by Dockerizing Spring Boot microservices with GitHub CI/CD pipelines.

PROJECTS

Chess Transformer Lichess Bot | PyTorch, ONNX, AWS EC2 | [sonuishaq67/chess](https://github.com/sonuishaq67/chess) | lichess/Q_K_V | Personal Project

- Trained a **500M**-param decoder-only transformer on **10B** tokens using PyTorch mixed precision across 8xGaudi2 GPUs.
- Built a data pipeline processing **100M+** games using DuckDB, UCI tokenization, and numpy memmaps.
- Served real-time inference at **sub-second** latency by exporting to ONNX and deploying on AWS EC2 via Lichess Bot API.

Pong Deep Q-Network Reinforcement Learning | PyTorch, Gymnasium, CNN | [sonuishaq67/pong](https://github.com/sonuishaq67/pong) | Personal Project

- Achieved **100%** win rate over 10K games by training a PyTorch Double DQN with experience replay and target updates.
- Accelerated training by **91%** by parallelizing RL across 16 AsyncVectorEnv workers on GPU cluster with torch.compile.

Amazon Reviews RAG System | Python, PostgreSQL, pgvector, Neo4j | Team Project

- Boosted precision@10 by **27%** across 571M Amazon reviews with a hybrid vector + knowledge graph retrieval pipeline.
- Quantified a **20%** precision gain from hybrid retrieval via ablation studies against vector-only and graph baselines.

Agent-Debate: Multi-Agent Research System | Python, OpenAI Agents, ElevenLabs, Beliefs SDK | [agent-debate](https://github.com/sonuishaq67/agent-debate) | Hackathon

- Built a multi-agent orchestrator chaining **5** agents (plan, retrieve, summarize, critic, write) for grounded reports.
- Synthesized debate podcasts with **ElevenLabs** TTS, tracking per-turn claim confidence via Beliefs SDK to resolve gaps.

ML-Driven Network Intrusion Detection System | Scikit-learn, XGBoost, Jupyter | Class Project

- Trained an XGBoost classifier on the NSL-KDD dataset to flag malicious network connections across various attack types.
- Cut the input space by 56% using correlation filtering and gain based importance, reducing training time with no accuracy loss.